

Image Segmentation using Thresholding for Cell Nuclei Detection of Colon Tissue

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Abstract—In this paper, image segmentation process using thresholding technique is discussed. We focus our study for nuclei detection of Hematoxilin Eosin (HE) stained colon tissue for the cancer detection. Detection of cancer cell at early stage is very important and several studies are being carried out. In medical imaging for the processing of microscopic tissue images and especially the detection of cell nuclei is done more and more using digital imagery techniques. However, there are many other techniques varies in complexity and quality of detection. In this paper the result of different thresholding techniques are applied on HE-stained colon tissue to detect cell nuclei in the image. In the proposed method, a binary mask is formed which is of the same size as that of original image. In this binary image, the centre of cell nuclei is represented by white colour and the background in black. This image is superimposed over the original image. The outcome of this is a segmented image to detect the cell nuclei. The process is relatively simple and results are encouraging.

Keywords—colon tissue; segmentation; thresholding; hernatoxilin eosin.

I. INTRODUCTION

Image segmentation [1] is the process of partitioning the digital image to detect and highlight different regions of interest in the image indicating multiple objects governed by unique features. With this method, one can modify the image to highlight the desired features or objects present in the image for further analysis. Using image processing techniques [2], the analysis and diagnosis is becoming more precise and less time consuming. With the advancements in the biomedical instruments, pathologists can acquire microscopic digital images (as shown in Fig.1). In a normal colon tissue the cells will be arranged in a ring like structure which makes glands [3]. Each of these cells will include a single nucleus with definite shape and the size. By detecting the cell nuclei one can inspect the shape, size and the architecture of the cell and able to distinguish between a normal cell and the cancerous cell. In the cancerous cells, the nuclei will be larger, and multiple nuclei can be found in a single cell. So in the case of colon cancer detection, it is important to detect the cell nuclei for further diagnostics.

There are many methods used for this purpose [3]. One of the important schemes used for image segmentation is seeded region growing algorithm [4]. In this approach, the initial seed

points selected are based on some criterion which differentiate them from the other regions and are given as input points [5]. Then the neighboring pixels having similar feature value are added to the original seed points based on some threshold and maximum area of the region of interest. The main task in this approach is to select the correct seed point. It is important and difficult as well. Furthermore, the amount of memory requirement is huge.

To overcome this problem, the input image is first split into smaller regions of equal size and saved as separately images [2]. After this, a region growing image segmentation is applied on these split images. Once the segmentation is completed, these split images are then merged again to get the original image.

Authors in [6], have transformed the hematoxilin eosin (HE) stained colon tissue color image from red green blue (RGB) space to Hue-saturation-value (HSV) color space after filtering. This results in formation of binary image. Morphological operations [7] are used to erode the image. This is then followed by maximum search and the linear component connect algorithm [8]. This algorithm is implemented partly on general purpose graphical processing unit (GPGPU) and sequential part on central processing unit (CPU).

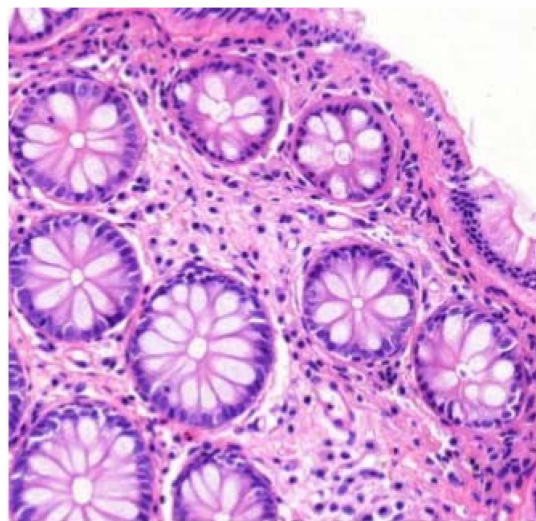


Fig. 1: HE-stained colon tissue

In the HE-stained colon tissue, the cell nuclei take a dark purple color which differentiates it well from the background. We can use threshold technique to segment the image. In our work, we have used multiple methods like segmentation using Sobel operator, Canny's method, Vector gradient method in RGB space, segmentation using pseudo coloring, segmentation in HSV color space. In all these approaches, we prepared the binary mask and examined it for the desired result. Here we are looking for a binary mask in which only the locations of the cell nuclei should be highlighted. Among all the above mentioned methods, the vector gradient method is detecting the edges of the cells properly whereas the HSV color space and thresholding method is correctly highlighting the locations of the nuclei in the image. In the proposed method, instead of using linear component connect algorithm we are fusing the original image with the binary image. In the resulting composite image the cell nuclei will be highlighted completing the segmentation process.

Rest of the contents in this paper is as follows. In section II, the overall operation is discussed. The detailed analysis of the proposed algorithm is presented in section III. The results are presented and discussed in Section IV. Section V concludes the paper with intended work in future.

II. OVERALL MECHANISM

Fig. 2 shows the work flow for segmenting image used in this work. Some of the steps are general which are common in many algorithms but in some of the processes, our approach is different which are discussed in section III. The steps are briefly explained below.

The input image is filtered using the Gaussian filter [7]. This blurs the input image reducing the unnecessary high frequency details. But still it retains the cell nuclei details. The threshold is found using Otsu's method [9]. The threshold is

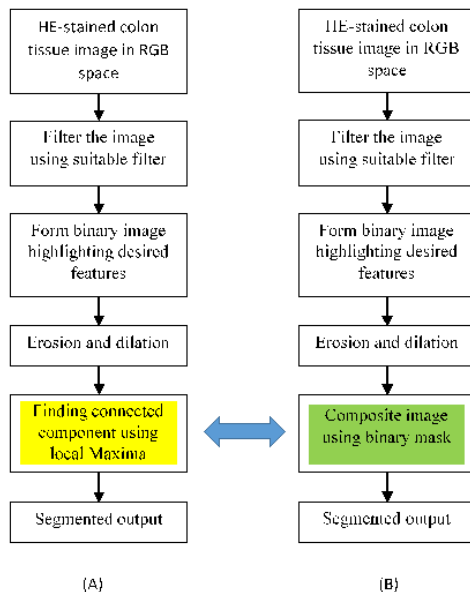


Fig. 2: Work flow for segmenting the image

found such that it minimizes the inter-class variance. Here the two classes are background pixels and foreground pixels. For converting the image into binary format, the filtered image is used. Forming the accurate binary image is the heart of the algorithm. Various methods are used in this regard and analyzed. The best result is then used for the next steps to segment the image by superimposing the binary mask on to the original image.

III. THE PROPOSED ALGORITHM AND INTERMEDIATE RESULTS

We discuss the proposed algorithm in this section. As discussed before, some of the steps remain common in many algorithms. But the kernel used for filtering the image or method used to convert the image into the binary form varies from image of one type to the image of another type. In Fig.2 the major change is highlighted. We explain each step in this section and also emphasize our approach and outcome of this.

A. Denoising the image

The HE-stained colon tissue image as shown in Fig.1 is first filtered using Gaussian filter. A 3x3 binomial approximation is used as mask to filter the noise. The Gaussian filter blurs the image but retains the color information of cell nuclei. This filter is convolved separately on the red, blue and green plane of the input image and the result is combined together to get the filtered image in RGB format again (Fig.3). The cell nuclei take a dark purple color which differentiates it well from the other parts of the tissue. The filtered image is then used for further processing. To prepare the binary mask, different methods are used (explained below) and the resulting images are given for visualization.

B. Segmentation using Sobel operator

In this method, first the input de-noised color image in RGB space is converted into gray scale image (Fig. 4). Then its binary gradient mask using Sobel operator [6] is calculated. By doing this, we tried to detect contours of the nuclei.

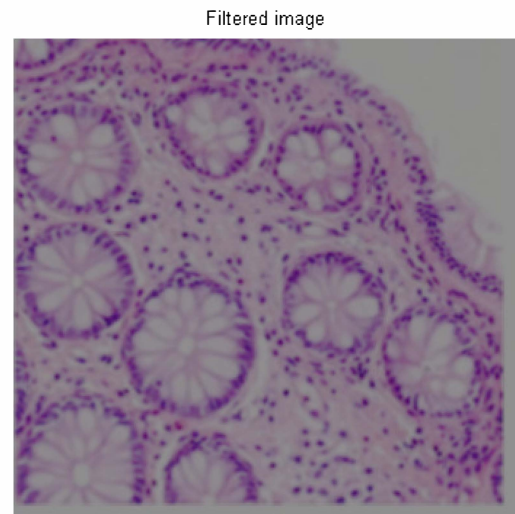


Fig. 3: Filtered image

Gray scale Image

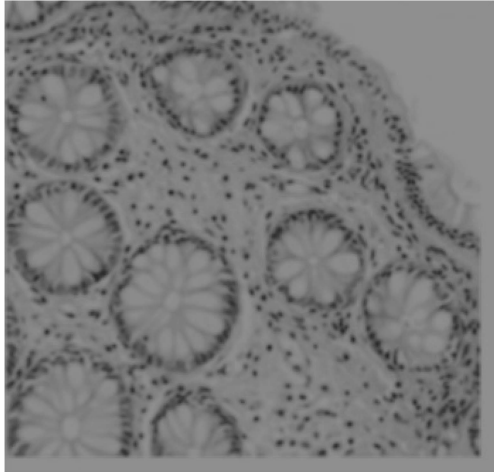


Fig. 4: Grayscale image

Binary image using sobel operator

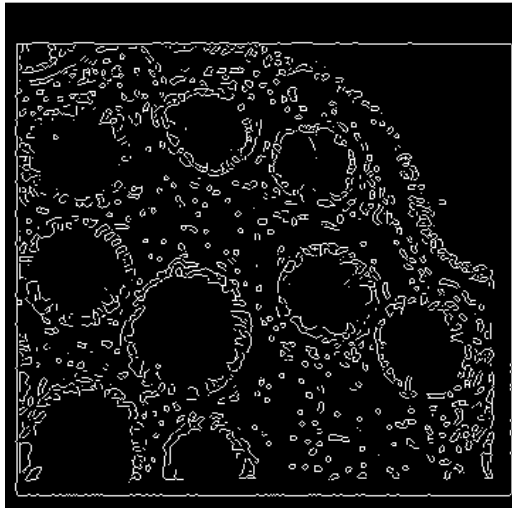


Fig. 5: Binary image using Sobel operator

segmetation using Canny operator



Fig. 6: Binary image using Canny's method

This gradient is then dilated using the structuring element as line at 45degree and length of 2. The holes are then filled followed by removing the borders. But this does not give the proper result only for identifying the nuclei. Other operator such as Prewitt also gives similar result (Fig.5).

C. Segmentation using Canny operator

The Canny method [6] finds edges by looking for local maxima of the gradient of the grayscale image. The gradient is calculated using Sobel operator after the image is filtered as shown in Fig.3. The gradient is found in four directions: horizontal, vertical and diagonal. This scheme uses two thresholds to detect the strong and the weak edges, and comprises the weak edges in the output only if they are linked to strong edges. The main advantage is that it gives single response to an edge and not multiple responses. By doing this, it is less likely to be misled by noise, and more likely to detect true weak edges. But it does not detect the borders of individual nuclei which are spaced very close to each other or which are connected to each other by weak edges (as observed in the resulting image Fig. 6). Due to this, the overlapped nuclei are not getting properly and separately detected.

D. Segmentation using Pseudocoloring

In this approach, the RGB image is first converted into grayscale image and the image contrast is increased by subtracting the uniform background from the original image. It is then converted into a binary image and a single element is searched which doesn't give correct result for this type of image. Then the binary image is pseudo colored. The resulting image is shown in Fig.7.

E. Vector gradient method

To find the binary gradient mask using Sobel operator on the gray scale image and only in one direction does not give very good result. So vector gradient method is used to find the gradient mask. In this method, the gradient in the RGB color space is computed. Assume a vector $C=[R \ G \ B]^T$ represents an arbitrary vector in RGB color space and r , g and b be the unit vector along R, G and B axes of RGB color space.

$$u = \left(\frac{\partial R}{\partial x}\right) r + \left(\frac{\partial R}{\partial x}\right) g + \left(\frac{\partial R}{\partial x}\right) b \quad (1)$$

pseudocolored image

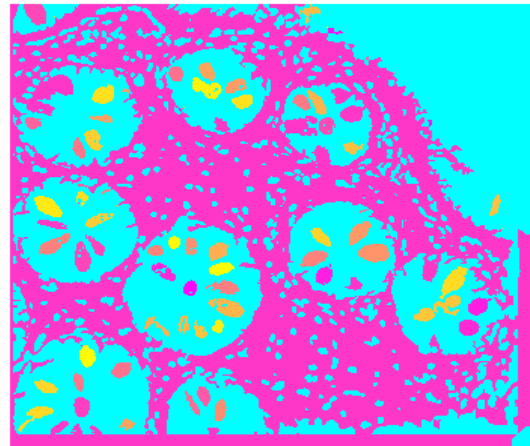


Fig. 7: Segmented image using Pseudo coloring

$$v = \left(\frac{\partial R}{\partial y}\right)r + \left(\frac{\partial R}{\partial y}\right)g + \left(\frac{\partial R}{\partial y}\right)b \quad (2)$$

The quantities g_{xx} , g_{yy} and g_{xy} are calculated as inner product of these vectors and given as:

$$g_{xx} = u^T u; \quad g_{yy} = v^T v; \quad g_{xy} = u^T v$$

Then the direction of maximum rate of change of $c(x,y)$ as a function of (x,y) is calculated by the angle given as:

$$\Theta(x,y) = \frac{1}{2} \tan^{-1} [2g_{xy} / (g_{xx} - g_{yy})] \quad (3)$$

Magnitude of the gradient in the direction given by the angle $\Theta(x,y)$ as:

$$F(x,y) = \left\{ \frac{1 [(g_{xx} + g_{yy}) + (g_{xx} - g_{yy}) \cos 2\Theta(x,y) + 2g_{xy} \sin 2\Theta(x,y)]}{2} \right\}^{1/2} \quad (4)$$

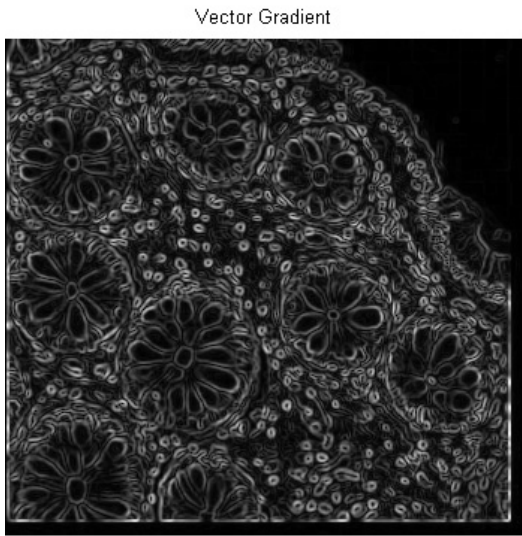


Fig. 8: vector gradient of RGB Image

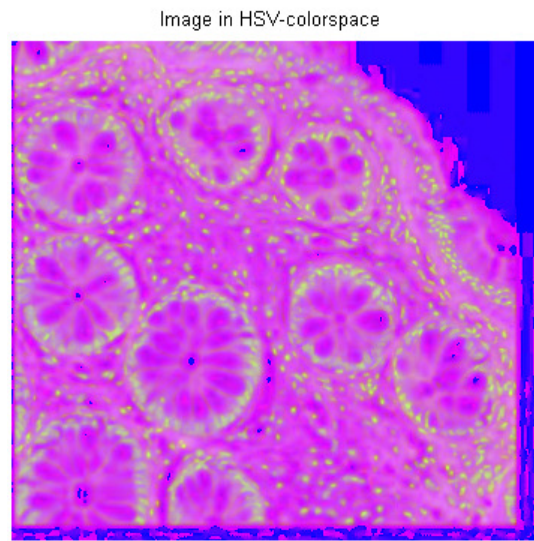


Fig. 10: Image transformed in HSV color space

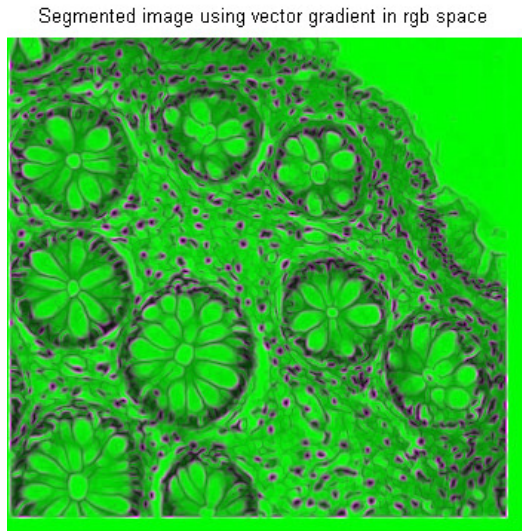


Fig. 9: Segmented image using vector gradient method

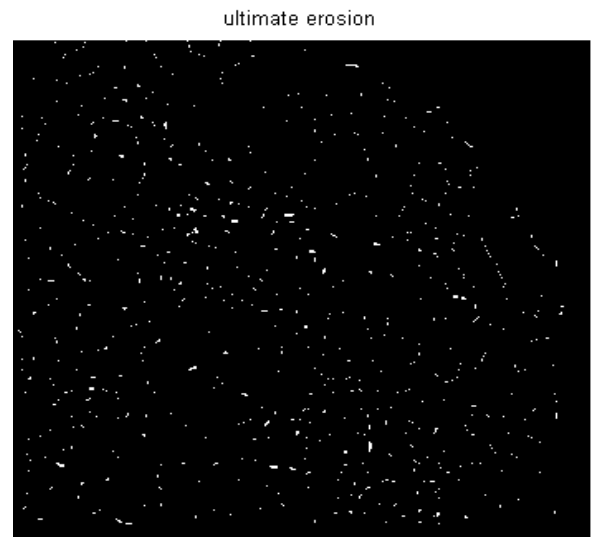


Fig.11: Ultimate Erosion

The resulting image is shown in Fig.8. It gives better result for the binary gradient mask. It detects the cell nuclei as well as cell boundaries properly which can be further used for studying the shape and the size of the cells present in the colon tissue. This gradient image is fused with the original image to highlight the cell nuclei (Fig.9).

F. Using HSV color space and Morphological operations

The filtered image is transformed from RGB to HSV color space (Fig.10) where it is easy to get the color information. This is then transformed into a binary image using Otsu's method to convert the image into binary form. This is then followed by ultimate erosion which gives the exact location of the cell nuclei (Fig.11).

The result is then used to make a composite image. In this, the regions showing nuclei are highlighted with magenta color and the background is of green color. The final image is shown in Fig.12.

Segmented Image

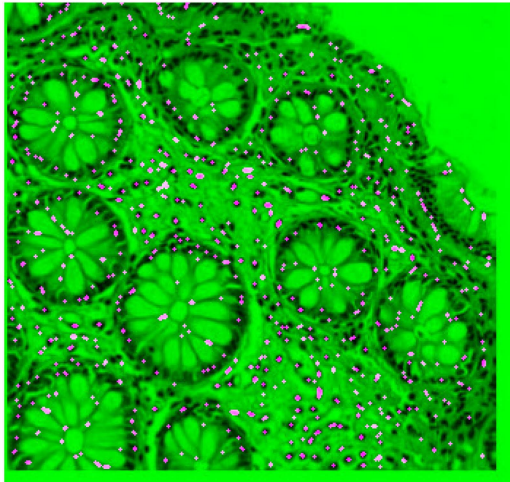


Fig 12: segmented image using vector gradient method

IV. DISCUSSIONS

The following images illustrate the resulting images obtained at various stages as mentioned above. In this work we have segmented the image to detect cell nuclei. We have applied this method on six different images of colon tissue. Due to lack of space we are showing the results only for one image in this paper. Since the cell nuclei in HE-stained colon tissue takes darker color as compared to the background, this feature is the main factor used to threshold the image and detect and highlight the cell nuclei. All the methods mentioned and described above, the segmentation using the HSV color space and segmentation using vector gradient method gave promising result. MATLAB computing environment is used to develop the algorithm and visualize the results.

V. CONCLUSION AND FUTURE WORK

Image segmentation techniques are ad-hoc based and differ usually from one type of the image to another based on the region and the properties of the region of interest. This implemented thresholding technique shows good quality of image detection.

As a future work we will find the cell nuclei concentration per unit cytoplasm area. This information is useful for cancer detection. The above mentioned methods can also be implemented using GPGPU to speed up the process. The only bottleneck is memory requirement for processing.

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